Mario Navarro Zea

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EDUCATION

Master of Science in Space Engineering ..... *[2023 – 2026 (expected)]*
Politecnico di Milano, Milan, Italy.

Key Modules: Space Propulsion, Combustion in Thermochemical Propulsion, Spacecraft Guidance and Navigation, Spacecraft Attitude Dynamics, Space Systems Simulation, PCB Design (Altium).

Master of Science in Aeronautical Engineering ..... *[2023 – 2026 (expected)]*
Universidad de Sevilla, Seville, Spain.

Key Modules: Helicopters, Turbomachinery Design, Spacecraft Systems, Flight Dynamics, Aeroelasticity, Composite Materials, Computational Fluid Dynamics.

Bachelor of Science in Aerospace Engineering ..... *[2019 – 2023]*
Universidad de Sevilla, Seville, Spain.

Final Thesis: "Comparative analysis of performances of different motor-propeller combinations for the characterization of electric motors efficiency". Work presented in the **20th PEGASUS Student Conference** (2024) and the **51st European Rotorcraft Forum** (2025).

Key Modules: Orbital Mechanics, Flight Mechanics, Fundamentals of Propulsion, Structures, Advanced Aerodynamics, CAD/CAM.

PROFESSIONAL EXPERIENCE

Intern in the Technology Department ..... *[May 2026 – current]*
European Space Agency, Redu, Belgium.

Development of a CubeSat Shock Testing Facility:

- Composition of the Assembly, Integration and Verification plan.
- Production of user manual and other documentation.
- Development of the test procedure.

Summer Engineering Intern ..... *[2022]*
Spanish Air and Space Force, Seville, Spain.

Study and analysis of **maintenance control** of aeronautical parts from *Spanish Air Force.*

- ***McDonell Douglas F-18 Hornet*:** analysis of maintenance records of parts from the pneumatic system.
- ***CASA CN-295:*** creation of maintenance and repair instructions of the main landing gear, **collaboratively** with the mechanics in order to make their work easier.
- ***CASA CN-235:*** design in CATIA V5 of an accessory for the propeller hub to be mounted onto its test fixture. Head of department had to place their **trust** in me with this project.

AWARDS

Best Bachelor's Thesis on an aerospace topic ..... *[Oct 16, 2024]*

By AIRBUS GROUP Aerospace Chair, Seville, Spain.

Award to the Best Bachelor's Thesis on an aerospace topic defended between 2019 and 2023 at ETSI - Higher Technical School of Engineering of the University of Seville, by the **AIRBUS GROUP Aerospace Chair**.

CONFERENCES

51st European Rotorcraft Forum ..... *[Sept 9 – 12, 2025]*

Venice, Italy.

Presented the characterization of electric motors efficiency for the development for eVTOL convertible aircraft. Professional visit to Leonardo's Final Assembly Line in Tessera.

20th PEGASUS Student Conference ..... *[Apr 26 – 27, 2024]*

Universitat Politècnica de Catalunya, Spain.

Student representative of the University of Seville, in its contributions to the PEGASUS European network in aerospace scientific research. Presented my Bachelor's Final Thesis.

INVOLVEMENT

Participant in the 2nd Electric Propulsion Summer School ..... *[Apr 1 – 3, 2025]*

ESTEC, Noordwijk, Netherlands.

Lectures on electric propulsion fundamentals, technologies, simulations, and mission applications. Networking with professionals, researchers, and students in the field of space electric propulsion.

Participant in the ESA Academy's Robotics Workshop ..... *[Oct 29 – Nov 1, 2024]*

ESTEC, Noordwijk, Netherlands.

Intensive 4-day workshop on space robotics. Worked developing **navigation algorithms** and **machine learning models** for object detection and autonomous exploration, applied to an ExoMy Rover, an open-source prototype inspired by ESA's Rosalind Franklin design.

Participant in the 22nd CVA Summer School ..... *[Jul 2 – 22, 2023]*

Vernon, Normandie, France.

Summer School organised by ***Community of Ariane Cities*** and hosted by educational institutions and supported by agencies, industrial firms and local sponsors, such as ***ArianeGroup*** **or** ***ESA.*** Based in space transportation and targeted at university students and young engineers from the aerospace industry, enhancing **international cooperation and inclusion**.

Intern at the Department of Aerospace Engineering ..... *[Dec 1, 2022 – Jun 30, 2023]*

Universidad de Sevilla, Sevilla.

Research for my Bachelor's Final Thesis within propulsive studies. Wind tunnel testing for the characterization of electric motors efficiency for the development of eVTOL convertible aircraft.

LANGUAGE SKILLS

English: ***Cambridge* Advanced English (Level C1),** grade B.

French: **DELF B2**, *Alliance Française*, 80/100.

German: **B1 level**, *EOI Spain (Official Language School).*

Italian: beginner level, from my Erasmus+.